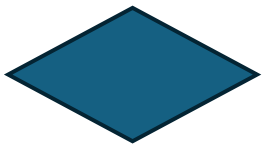
## System Flow Diagram :-

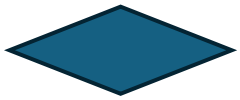
A System Flow Diagram (CFD) is a visual representation that illustrates the logical sequence of processes and decision-making within a system. It begins with an entry point, such as user login or system initialization, and ends with an exit point, like a successful transaction or process completion. Tasks or operations are represented by rectangles, while decisions are depicted as diamonds to indicate conditional branching based on specific criteria. Arrows connect these components, showing the flow of control from one step to another.

| SYMBOL | NAME | DESCRIPTION |
|---|---|---|
| | Process / Task | Indicates a specific operation, activity, or task performed in the system. |
|  | Decision Point | Indicates a conditional branch in the control flow. The control follows one of the paths based on the outcome of the decision |
| | Control Flow | Shows the direction of the control flow from one task to the next. |

# Activity Diagram :-

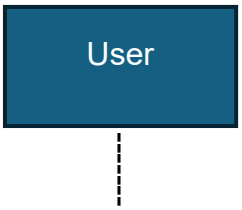
An Activity Diagram is a visual representation of the workflow or sequence of activities within a system, showcasing how tasks are executed and flow from one to the next. It highlights the dynamic behavior of the system by illustrating actions, decisions, loops, and parallel processes. Activity diagrams use symbols like ovals for activities, diamonds for decision points, and arrows to indicate the flow of control. Swimlanes may be used to assign responsibilities to specific actors or system components.

| Symbol | Name | Description |
|---|---|---|
| | Initial State | A black circle is standard notation for initial state before an activity take place. It can either stand alone or can be further elucidated using a note. |
| | Action State | The activity symbols are the baisic building blocks of an activity diagram and usually have a short description of the activity they represent. |
| | Control Flow | Arrows represent the direction flow of the flow chart. The arrow points in the |

| | | |
|---|---|---|
| | | direction of the Action State Page \| 34 progressing activities. |
|  | Decision | A marker shaped like a diamond is the stand symbol for a decision. |
| | Final State | The black circle that looks like a selected radio button is the symbol for end of an activity |

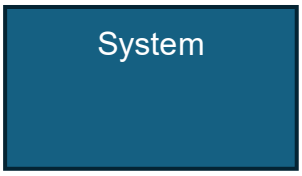
# Sequence Diagrams :-

A Sequence Diagram illustrates how system components or objects interact in a specific order to complete a task. It shows the flow of messages or events between actors and objects over time, using vertical lifelines and horizontal arrows.

| Symbol | Name | Description |
|---|---|---|
|  | Timelines | A Timeline represents an individual participant in a sequence diagram. A lifeline will usually have a rectangle containing its object name. |
| | Focus of Control | Focus of control is used in sequence diagram to show the period of time during which an object performed an action. And also show the **Sequence** of Action. |

## Use Case :-

A Use Case is a detailed description of how users interact with a system to achieve specific goals. It represents a sequence of actions or events that a system performs in response to a user's input, focusing on what the system does rather than how it is implemented. Use cases are typically represented using diagrams and narratives that include the primary actor (user or another system), the system itself, and the interaction between them. A Use Case Diagram visually depicts these interactions, with actors represented as stick figures and use cases as ovals connected by lines to show relationships.

| Symbols | Name | Description |
|---|---|---|
| | Actor | Represents a user or external system interacting with the system. Examples: "Customer," "Admin." |
| | Entity | Describes a specific functionality or interaction. Examples: "brute force," "Input Pass" |
|   | System | Represents the scope of the system, enclosing all use cases that belong to it. |

| | | |
|---|---|---|
| | Connection | Indicates the interaction between actors Page \| 28 and use cases or between use cases themselves. |
| <<include> | Include | Represents **mandatory** functionality that a use case includes. |
| <<Extend>> | Extend | Represents **optional** functionality extending a use case. |